Test Automation Framework (TAF::Perl)

DATABASE ITERATION MODES AND TEST SETUP MODES

MariaDB Foundation https://mariadb.org

Open, vendor-neutral, community-driven tooling for database testing and benchmarking.

TAF-Perl is maintained and published by the MariaDB Foundation to support transparent, reproducible, and fair performance evaluation across the open source database ecosystem. The framework is part of the Foundation's mission to provide open tooling that benefits the entire database community.

TAF DATABASE ITERATION MODES AND TEST SETUP MODES

CheckTestSetup decides two things before each iteration:

1. What to do with the database (preserve, restart, reset, restore)
2. Whether to run MainTestSetup (skip, once, per_test, per_iter, or forced by reset/restore)

It uses three pieces of state:

- first_time_in_tests_loop
- initial_test_setup_done
- restore_created

And two user options:

- database_iteration_mode
- test_setup_mode

Below is the operational behavior.

FIRST ITERATION

On iteration 1:

- No restart/reset/restore happens yet.
- Test setup mode decides whether MainTestSetup runs, except in reset or restore modes.
- Warmup runs only if enabled and not already done.

This guarantees a clean and predictable first iteration.

RESTORE MODE

database_iteration_mode = restore

Iteration 1:

- Always run MainTestSetup.
- Stop DB.
- Create the restore image.
- Start DB.
- Warmup always runs after restore if warmups were enabled.

Iteration 2 and later:

- Stop DB.
- Restore datadir from the image.
- Start DB.
- Warmup always runs after restore if warmups were enabled.

Restore mode gives identical database state every iteration with minimal cost.

RESET MODE

database_iteration_mode = reset

- Stop DB.
- Recreate database from scratch.
- Start DB.
- Always run MainTestSetup.
- Warmup always runs after reset if warmups were enabled.

Reset mode gives a fresh database every iteration.

RESTART MODE

database_iteration_mode = restart

- Restart DB before each iteration.
- Warmup always runs after restart if warmups were enabled.
- MainTestSetup runs only if test_setup_mode requires it.

Restart mode gives a cold engine but stable data.

PRESERVE MODE

database_iteration_mode = preserve

- DB is left running.
- MainTestSetup runs only if test_setup_mode requires it.
- Warmup usually runs once unless manually reset.

Preserve mode is the fastest and cheapest.

TEST SETUP MODE

test_setup_mode = skip once per_test per_iter

- skip: Never run MainTestSetup.
- once: Run only on the first iteration.
- per_test: Run once per test block.
- per_iter: Run on every iteration.

Important: If database_iteration_mode = restore, test_setup_mode is ignored. Restore mode always runs MainTestSetup on iteration 1 and never again.

WARMUP RULES

Warmup runs when:

- warmups are enabled (include_warmup_iteration)
- warmup_run_done = FALSE

Warmup_run_done is reset whenever DB state changes:

- restart
- reset
- restore

Therefore: Warmup always runs after restore, restart, or reset if warmups were enabled.

RECOMMENDED SETTINGS

- Fastest runs: database_iteration_mode = preserve test_setup_mode = once
- Cold engine, stable data: database_iteration_mode = restart test_setup_mode = once
- Fresh database every iteration: database_iteration_mode = reset
- Identical database state every iteration: database_iteration_mode = restore restore_image_format = copy | tar | tar.gz

End of Document

This Quick Start was prepared and published by the MariaDB Foundation https://mariadb.org

TAF::Perl is part of the Foundation's commitment to open, vendor-neutral tooling for the database ecosystem. The framework is maintained to support fair benchmarking, reproducible testing, and transparent engineering practices across all database makers.

For contributions, feedback, or questions, please visit the Foundation website or contact the maintainers directly.